

Practical 7th

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user. i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

CODE :

```
#include <stdio.h>
int main()
{
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int frame[3], time[3];
    int i,j,k,flag;
    int pageFault = 0, pageHit = 0, counter = 0;
    for(i=0;i<frames;i++)
    {
        frame[i] = -1;
        time[i] = 0;
    }
    for(i=0;i<n;i++)
    {
        flag = 0;
        for(j=0;j<frames;j++)
        {
            if(frame[j] == pages[i])
            {
                counter++;
                time[j] = counter;
                flag = 1;
                pageHit++;
                break;
            }
        }
    }
}
```

```

    }
    if(flag == 0)
    {
        int lru = 0;
        for(k=1;k<frames;k++)
        {
            if(time[k] < time[lru])
                lru = k;
        }
        frame[lru] = pages[i];
        counter++;
        time[lru] = counter;
        pageFault++;
    }
}

printf("Page Hits = %d\n",pageHit);
printf("Page Faults = %d\n",pageFault);

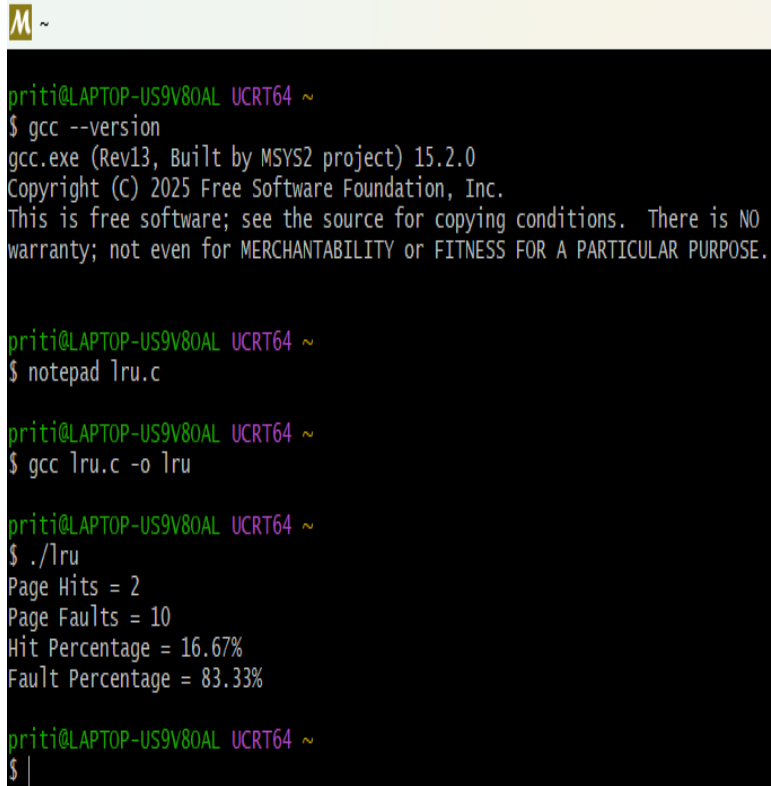
float hitPercent = ((float)pageHit/n)*100;
float faultPercent = ((float)pageFault/n)*100;

printf("Hit Percentage = %.2f%%\n",hitPercent);
printf("Fault Percentage = %.2f%%\n",faultPercent);

return 0;
}

```

OUTPUT :



```
M ~
priti@LAPTOP-US9V80AL UCRT64 ~
$ gcc --version
gcc.exe (Rev13, Built by MSYS2 project) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

priti@LAPTOP-US9V80AL UCRT64 ~
$ notepad lru.c

priti@LAPTOP-US9V80AL UCRT64 ~
$ gcc lru.c -o lru

priti@LAPTOP-US9V80AL UCRT64 ~
$ ./lru
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

priti@LAPTOP-US9V80AL UCRT64 ~
$ |
```